Microeconomics 2

Economics and Data Science 244

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Problem Set #1 - Perfect Competition and Monopoly

Q1 (20 points) Competitive Equilibrium

Consider a market with two consumers (indexed by $i = 1, 2$) and two firms (indexed by $j = 1, 2$). Consumers' demand functions are given by

$$D_1(p) = 10 - p \qquad \text{and} \qquad D_2(p) = 13 - 0.5p,$$

and the firms' cost functions are given by

$$C_1(q) = \begin{cases} 0 & q = 0 \\ 8 + \frac{q^2}{2} & q > 0 \end{cases} \qquad \text{and} \qquad C_2(q) = 2q + q^2 \quad \forall q \geq 0.$$

1. (a) (5 points) Construct the market demand function $D(p)$ and the inverse market demand function $p^D(q)$.
2. (b) (5 points) Construct the market supply function $S(p)$ and the inverse market supply function $p^S(q)$.
3. (c) (5 points) Find the competitive equilibrium (both quantity and price) and determine individual supply and demand quantities.
4. (d) (5 points) Calculate consumer surplus for each consumer and producer surplus and profit for each firm.

Given:

- $D_1(p) = 10 - p$
- $D_2(p) = 13 - \frac{1}{2}p$
- $C_1(q) = \begin{cases} 0 & q = 0 \\ 8 + q^2/2 & q > 0 \end{cases}$
- $C_2(q) = 2q + q^2 \quad \forall q \geq 0$

(a) Market Demand and Inverse Market Demand

> **Note**
>
> Market demand is the sum of individual demands, but we mustn't forget that each will start consuming at a different price.

$$D_1(p) + D_2(p) = (10 - p) + \left(13 - \frac{1}{2}p\right) = 23 - \frac{3}{2}p$$

$$D(p) = \begin{cases} 0 & p \geq 26 \\ 13 - \frac{1}{2}q & 10 \leq p < 26 \\ 23 + \frac{3}{2}q & p < 10 \end{cases}$$

Inverse market demand function $p^D(q)$:

$$q = 23 - \frac{3}{2}p \Rightarrow \frac{3}{2}p = 23 - q \Rightarrow p^D(q) = \frac{2(23 - q)}{3}$$

$$q = 13 - \frac{1}{2}p \Rightarrow p = 26 - 2q$$

✎ Note

The inverse demand function also gets certain limitations placed on q.

We obtain the inverse demand function from their respective demand function, and the intervals for each section are found by substituting the values of the intervals for the price of the demand function into their respective demand functions.

> (1) $13 - \frac{1}{2}(10) = 8$, as an example of how it was done.

$$p^D(q) = \begin{cases} \frac{46-2q}{3} & 8 \leq q \leq 23, \\ 26 - 2q & 0 \leq q < 8. \end{cases}$$

(b) Market Supply and Inverse Market Supply

Firstly, we look for the shutdown price.

$$p = MC \geq AVC(q) + \frac{F - S}{q}$$

✎ Note

$p = MC$ seems pretty self explanatory at this point. Nothing would get produced if the marginal costs outpace the price of the produced good. The second section of the condition comes in.

The $p \geq AVC(q)$ is the condition for the firm to continue operating, if $p < \min AVC$ shutdown happens. On top of that we consider F - the entrance fee into the marke, and S - the sunk costs, for each of the sold items (q).

Thus.

For AVC_1

$$q \geq \frac{q}{2} + \frac{8}{q}$$

$$q \geq \frac{8}{q} + \frac{q}{2}$$

$$\frac{2q^2 - 16 - q^2}{2q} \geq 0$$

$$\frac{q^2 - 16}{2q} \geq 0 \Rightarrow q > 4$$

For AVC_2

$$\frac{2q + q^2}{q} = 2 + q \Rightarrow q > 2$$

Individual supply functions:

$$S_1(p) = \begin{cases} 0 & p < 4 \\ p & p \geq 4 \end{cases}$$

$$S_2(p) = \begin{cases} 0 & p < 2 \\ \frac{p-2}{2} & p \geq 2 \end{cases}$$

Market supply $S(p)$:

$$S(p) = S_1(p) + S_2(p) = p + \frac{p-2}{2} = \frac{3p-2}{2}$$

$$S(p) = \begin{cases} 0 & p < 2 \\ \frac{p-2}{2} & 2 \le p \le 4 \\ \frac{3p-2}{2} & p \ge 4 \end{cases}$$

Inverse market supply $p^S(q)$:

$$q = \frac{3p-2}{2} \Rightarrow 2q = 3p - 2 \Rightarrow 3p = 2q + 2 \Rightarrow p^S(q) = \frac{2q+2}{3}$$

> **Note**
>
> Once more, in order to find the new intervals for the inverse function, we must use the intervals from the original function and substitute them into their corresponding functions.

$$p^S(q) = \begin{cases} 2q + 2 & 0 \le q < 1 \\ \frac{2q+2}{3} & 5 \le q \end{cases}$$

(c) Competitive Equilibrium

> **Note**
>
> Competitive equilibrium allows us to set set $D(p) = S(p)$.

$$23 - \frac{3}{2}p = \frac{3p-2}{2}$$

$$46 - 3p = 3p - 2 \Rightarrow 48 = 6p \Rightarrow p^* = 8$$

$$q^* = D(8) = 23 - \frac{3}{2}(8) = 23 - 12 = 11$$

Individual quantities:

- $q_1^s = S_1(8) = 8$
- $q_2^s = S_2(8) = \frac{8-2}{2} = 3$
- $q_1^d = D_1(8) = 10 - 8 = 2$
- $q_2^d = D_2(8) = 13 - \frac{1}{2}(8) = 13 - 4 = 9$

(d) Surplus and Profit Calculations

Consumer Surplus:

> **Note**
>
> CS can be defined as the difference between what they are willing to pay and what they actually pay. Geometrically it is the area under the demand curve and above the equilibrium price.

> **Note**
>
> We find the equilibrium price p^* (for consumer 1 it is 2) using the equilibrium quantity we've already gotten. Then we can notice that the area of CS is a triangle, we can use the geometric formula for.

- $CS_1 = \frac{1}{2}(10-8)(2) = 2$

- $CS_2 = \frac{1}{2}(26-8)(9) = \frac{1}{2}(18)(9) = 81$

Producer Surplus:

> **Note**
>
> The PS can be defined as the difference between the revenue and the variable cost. It excludes fixed and start-up costs (thus $PS \geq \pi$).

- $PS_1 = \frac{1}{2}(8)(8) = 32$
- $PS_2 = \frac{1}{2}(8-2)(3) = 9$

Profit:

> **Note**
>
> $\Pi = TR - TC$

- $\pi_1 = TR_1 - TC_1 = (8)(8) - \left(8 + \frac{8^2}{2}\right) = 64 - (8+32) = 24$
- $\pi_2 = TR_2 - TC_2 = (8)(3) - (2(3) + 3^2) = 24 - (6+9) = 9$

Q2 (20 points) Monopsony Problem

Consider a market with two competitive suppliers indexed by $i = 1, 2$ and one monopsony firm on the demand side. The cost functions of suppliers are given by

$$C_1(q) = 2q + 2q^2 \qquad \text{and} \qquad C_2(q) = 4q + q^2.$$

The monetary value function of the monopsony is given by

$$V(q) = \begin{cases} 0 & q = 0 \\ 33q - \frac{q^2}{3}\left(4 - \frac{7}{q}\right) & q \in (0,9) \\ 210 & q \geq 9 \end{cases}$$

1. (a) (5 points) Find the supply function of each supplier, the market supply and the inverse market supply.
2. (b) (5 points) Solve for the socially optimal quantity, price, and individual outputs of suppliers.
3. (c) (5 points) Find the buying cost function and the marginal buying cost function of the monopsony.
4. (d) (5 points) Find the monopsony optimal price and quantity and determine individual quantities bought by monopsony from each supplier.

Given:

- $C_1(q) = 2q + 2q^2$
- $C_2(q) = 4q + q^2$
- $V(q) = \begin{cases} 0 & q = 0 \\ 33q - \frac{q^2}{3}(4 - \frac{7}{q}) & q \in (0,9) \\ 210 & q \geq 9 \end{cases}$

(a) Supply Functions and Inverse Market Supply

Firm 1: $MC_1 = 2 + 4q$.
Supply: $p = 2 + 4q$

$$S_1(p) = \frac{p-2}{4} : p \geq 2$$

Firm 2: $MC_2 = 4 + 2q$.
Supply: $p = 4 + 2$

$$S_2(p) = \frac{p-4}{2} : p \geq 4$$

Market Supply $S(p)$.
If both produce, then. $S(p) = \frac{p-2}{4} + \frac{p-4}{2} = \frac{3p-10}{4}$

$$S(p) = \begin{cases} 0 & 0 < 2 \\ \frac{p-2}{4} & p \in [2, 4) \\ \frac{3p-10}{4} & p \geq 4 \end{cases}$$

Inverse Market Supply $p^S(q)$:

> **Note**
>
> We must consider that at $p = 4$, $q_1 = \frac{4-2}{4} = \frac{1}{2}$ and thus the market $q = \frac{1}{2}$. Essentially, if we were having intervals on the price, we inverse it, and thus must have intervals on the quantity.

For $q \leq \frac{1}{2}$ (only Firm 1): $q = \frac{p-2}{4} \Rightarrow p = 2 + 4q$

For $q > \frac{1}{2}$ (both firms): $q = \frac{3p-10}{4} \Rightarrow p = \frac{4q+10}{3}$

$$p^S(q) = \begin{cases} 4q + 2 & q < \frac{1}{2} \\ \frac{4q+10}{3} & q \geq \frac{1}{2} \end{cases}$$

(b) Socially Optimal Quantity and Price

> **Note**
>
> We simplify the monopsony's value function for $0 < q < 9$:
>
> $$V(q) = 33q - \frac{4}{3}q^2 + \frac{7}{3}q = \frac{106}{3}q - \frac{4}{3}q^2$$

Marginal value (MV):

$$MV(q) = V'(q) = \frac{106}{3} - \frac{8}{3}q$$

> **Note**
>
> Social optimum: $MV(q) = MC(q)$, where MC is market marginal cost (inverse supply).

For $q > \frac{1}{2}$: $MC(q) = p^S(q) = \frac{4q+10}{3}$.

Now we solve for q^*.

$$\frac{106}{3} - \frac{8}{3}q = \frac{4q+10}{3}$$

$$106 - 8q = 4q + 10$$

$$96 = 12q$$

$$q^* = 8$$

Price:

$$p^* = p^S(8) = \frac{4(8)+10}{3} = \frac{42}{3} = 14$$

Individual outputs at $p^* = 14$:

$$q_1^* = S_1(14) = \frac{14-2}{4} = 3$$

$$q_2^* = S_2(14) = \frac{14-4}{2} = 5$$

c) Buying Cost and Marginal Buying Cost

$$\text{NBC}(q) = p(q) + p'(q)q = \frac{4q+10}{3} + \frac{4}{3}q = \frac{8q+10}{3}$$

$$\Rightarrow BC = \int \text{NBC}(q)dq = \int \frac{8q+10}{3}dq = \frac{4q^2}{3} + \frac{10q}{3}$$

$$\text{NBC}(q) = p(q) + p'(q)q = (4q+2) + 4q = 8q+2$$

$$\Rightarrow BC = \int \text{NBC}(q)dq = \int (8q+2)dq = 4q^2 + 2q$$

d) Find q^m (monopoly quantity)

Given inverse demand function. $p(q) = \frac{106-8q}{3}$

Marginal cost (MC) appears to be: $\frac{2q+10}{3}$

Set MR = MC:

$$\frac{d}{dq}[p(q) \cdot q] = \frac{d}{dq}\left[\frac{106q - 8q^2}{3}\right] = \frac{106-16q}{3}$$

$$\frac{106-16q}{3} = \frac{2q+10}{3}$$

$$106 - 16q = 2q + 10$$

$$96 = 18q$$

$$q^m = \frac{96}{18} = \frac{16}{3}$$

Find monopoly price:

$$p^m = \frac{106 - 8 \cdot \frac{16}{3}}{3} = \frac{106 - \frac{128}{3}}{3} = \frac{\frac{190}{3}}{3} = \frac{190}{9}$$

$$S_1(p) = \begin{cases} 0, & p < 2 \\ \frac{p-2}{2}, & p \geq 2 \end{cases}$$

At $p = \frac{190}{9}$:

$$q_1 = \frac{\frac{190}{9} - 2}{2} = \frac{\frac{190}{9} - \frac{18}{9}}{2} = \frac{\frac{172}{9}}{2} = \frac{86}{9}$$

$$S_2(p) = \begin{cases} 0, & p < 4 \\ \frac{p-4}{4}, & p \geq 4 \end{cases}$$

At $p = \frac{190}{9}$.

$$q_2 = \frac{\frac{190}{9} - 4}{4} = \frac{\frac{190}{9} - \frac{36}{9}}{4} = \frac{\frac{154}{9}}{4} = \frac{77}{18}$$

Total quantity: $q_1 + q_2 = \frac{86}{9} + \frac{77}{18} = \frac{172}{18} + \frac{77}{18} = \frac{249}{18}$

Q3 (40 points) Monopoly and Price Discrimination

A monopoly firm operates in a market with two consumers. The inverse demand functions of these consumers are given by

$$p_1(q) = 10 - \frac{q}{2} \qquad \text{and} \qquad p_2(q) = 8 - \frac{q}{2}.$$

The total cost function of the monopoly is given by

$$C(q) = 5q.$$

Answer the following questions:

1. (a) (10 points) Construct the market demand $D(p)$, the inverse market demand $p(q)$ and the marginal revenue function for the monopoly.
2. (b) (7 points) Solve for the monopoly outcome when it is limited to a single linear price (calculate quantity, price, profit).
3. (c) (7 points) Solve for the monopoly outcome under the first-degree price discrimination (find optimal tariffs and monopoly's profit).

For the rest of the question, suppose that the monopoly is allowed to use a **single two-part tariff**

$$T(q) = F + pq$$

where $F \geq 0$ is a fixed fee and $p \geq 0$ is a per unit price.

4. (d) (10 points) First, solve for the monopoly outcome assuming that both consumers have to be served (parameters of the optimal tariff, quantity, profit).
5. (e) (6 points) Now derive the optimal two-part tariff for the monopoly (parameters F and p of the optimal tariff, quantity, profit) taking into account that the monopoly can always choose to serve just one consumer.

Given:

- $p_1(q) = 10 - \frac{q}{2}$
- $p_2(q) = 8 - \frac{q}{2}$
- $C(q) = 5q$

(a) Market Demand, Inverse Demand, and Marginal Revenue

Individual demand functions from inverse demands:
$q_1 = 20 - 2p : p \leq 10$
$q_2 = 16 - 2p : p \leq 8$

Market Demand $D(p)$:

$$D(p) = \begin{cases} 0 & p > 10 \\ 20 - 2p & 8 < p \geq 10 \\ 36 - 4p & p \leq 8 \end{cases}$$

Inverse Market Demand $p(q)$:

Note

We can obtain the inverse marked demands from the individual market demands, which is what we did here. The intervals for q are born from the intervals of D(q), that we substitute into their respective functions.

$$p(q) = \begin{cases} 10 - \frac{q}{2} & 0 \leq q < 4 \\ 9 - \frac{q}{4} & q \geq 4 \end{cases}$$

Marginal Revenue $MR(q)$:

$$MR = \begin{cases} 10 - q & 0 \leq q < 4 \\ 9 - \frac{q}{2} & q \geq 4 \end{cases}$$

(b) Monopoly Outcome with Single Linear Price

Marginal cost: $MC = 5$.
Solve $MR = MC$.

Case 1: $p \leq 8$ (Both consumers buy)

$$\pi(p) = (p-5)(36-4p) = -4p^2 + 56p - 180 \to \max$$

First Order Condition (FOC).

$$\frac{d\pi}{dp} = -8p + 56 = 0$$

Second Order Condition (SOC). $\frac{d^2\pi}{dp^2} = -8 < 0$

Price.

$$-8p + 56 = 0 \Rightarrow p = 7$$

Quantity.

$$q = 36 - 4(7) = 36 - 28 = 8$$

Profit:

$$\pi(7) = (7-5)(36-28) = 2 \times 8 = 16$$

Case 2: $8 < p \leq 10$ (Only Consumer 1 buys)

Profit function:

$$\pi(p) = (p-5)(20-2p) = -2p^2 + 30p - 100 \to \max$$

First Order Condition (FOC).

$$\frac{d\pi}{dp} = -4p + 30 = 0$$

Second Order Condition (SOC). $\frac{d^2\pi}{dp^2} = -4 < 0$

Price.

$$-4p + 30 = 0 \Rightarrow p = 7.5$$

Quantity.

$$q = 20 - 2(7.5) = 20 - 15 = 5$$

Profit:

$$\pi(7.5) = (7.5-5)(20-15) = 2.5 \cdot 5 = 12.5$$

Thus.

Profit Comparison:
Case 1 ($p = 7$): $\pi = 16$
Case 2 ($p = 7.5$): $\pi = 12.5$

Optimal Monopoly Outcome:
Price: $p^m = 7$
Quantity: $q^m = 8$
Profit: $\pi^m = 16$

(c) First-Degree Price Discrimination

> **Note**
>
> We are working in $T(q) = F + p \cdot q$.

Set $p = MC = 5$ for both consumers and extract all surplus.

Consumer 1:
$q_1 = 20 - 2(5) = 10$
$\text{CS}_1 = \frac{1}{2}(10 - 5)(10) = 25$
Monopoly extracts CS + cost = $25 + 50 = 75$

Consumer 2:
$q_2 = 16 - 2(5) = 6$
$\text{CS}_2 = \frac{1}{2}(8 - 5)(6) = 9$
Monopoly extracts CS + cost = $9 + 30 = 39$

> **Warning**
>
> Here I got kind lucky, it is not always we get triangles that cover the area of the consumer surplus, so this is likely not the safest method.

Total revenue = $75 + 39 = 114$
Total cost = $5 \cdot 16 = 80$
Profit:

$$\pi = 114 - 80 = 34$$

$$t_1 = 25 + 5q, \quad t_2 = 9 + 5q$$

> **Note**
>
> The tariff goes on top of the purchased product.

(d) Two-Part Tariff (Serving Both Consumers)

Problem Setup:

> **Note**

We have a monopoly facing two consumers with different demand functions, using a two-part tariff $T(q) = F + p \cdot q$ where both consumers must be served.

Market Demand:

$$D(p) = \begin{cases} 0, & p > 10 \\ 36 - 4p, & p \leq 8 \end{cases}$$

Price Range: $p \in [0, 8]$ (to ensure both consumers are served)

Consumer 2
Demand.
$q_2 = 16 - 2p$
Consumer Surplus & fixed fee.
$CS_2(p) = \frac{1}{2}(8 - p)(16 - 2p) = (8 - p)^2$
$F = CS_2(p) = (8 - p)^2$

Total profit consists of.

- Variable profit: $(p - 5)(36 - 4p)$
- Fixed fees: $2F = 2(8 - p)^2$

Thus.

$$\Pi(p) = (p - 5)(36 - 4p) + 2(8 - p)^2$$

Expand the profit function:

$$\Pi(p) = (p - 5)(36 - 4p) + 2(8 - p)^2$$
$$= -2p^2 + 24p - 52$$

FOC:

$$\frac{\delta\Pi(p)}{\delta p} = -4p + 24 = 0$$

Price:

$$-4p + 24 = 0 \Rightarrow p = 6$$

Price: $p^* = 6$
Consumer 2's quantity: $q_2^* = 16 - 2(6) = 4$
Fixed fee: $F^* = CS_2(6) = (8 - 6)^2 = 4$
Total demand: $D(6) = 36 - 4(6) = 12$

Thus.

$$\Pi^* = (6 - 5) \cdot 12 + 2 \cdot 4 = 1 \cdot 12 + 8 = 20$$

Final Optimal Tariff:

- Per-unit price: $p^* = 6$
- Fixed fee: $F^* = 4$
- Total profit: $\Pi^* = 20$

(e) Optimal Two-Part Tariff (Monopoly May Serve One)

Consumer 1:
$q_1 = 20 - 2p$

Set $p = MC = 5$ (efficient pricing)
$q_1 = 20 - 2(5) = 10$
$CS_1(5) = (10 - 5)^2 = 25$
Set $F = CS_1 = 25$
$\pi_1 = F = 25$

Consumer 2
$p = 5$: $CS_2(5) = (8 - 5)^2 = 9$
To serve both, need $F \leq CS_2 = 9$
But $9 \leq 25$ constraint is too restrictive
$F = 9$: $\pi_{both} = 2(9) = 18$

Serve only Consumer 1: $\pi = 25$
Serve both consumers: $\pi = 18$
$25 > 18 \Rightarrow$ Optimal to serve only Consumer 1

Optimal tariff parameters
Per-unit price: $p^* = 5$
Fixed fee: $F^* = 25$
Quantity: $q^* = 10$
Profit: $\pi^* = 25$

Optimal Tariff:

$$T(q) = 25 + 5q$$

Q4 (20 points) Bundling Problem

Consider a market with two goods, A and B, and consumers of two types, Type 1 and Type 2. The number of Type 1 consumers is x, and the number of Type 2 consumers is y, where

$$x \geq y \geq 1.$$

Each consumer has an independent unit demand for each good with the following willingness to pay:

	A	B
Type 1	10	10
Type 2	12	4

The monopoly has unlimited supply of each good.

1. (a) (5 points) Find the firm's highest profit from selling both goods separately (i.e., individual pricing).
2. (b) (5 points) Find the firm's highest profit from pure bundling.
3. (c) (10 points) For all possible values of (x, y), determine the best pricing regime for the monopoly.

Given:

- Type 1: WTP $A = 10$, $B = 10$
- Type 2: WTP $A = 12$, $B = 4$
- $x \geq y \geq 1$

(a) Highest Profit from Separate Selling

> **Note**
>
> For each good, the monopolist faces a trade-off: set a low price to sell to both types or a high price to sell only to the type with the higher valuation.

Good A:
Price can be 10 (sells to both) or 12 (sells only to Type 2).

$$p_A = 10 \Rightarrow \pi_A = 10(x+y)$$
$$p_A = 12 \Rightarrow \pi_A = 12y$$

Since $x \geq y$, $10(x+y) \geq 10(2y) = 20y > 12y$. We will set $p_A = 10$, $\pi_A = 10(x+y)$

Good B:
Price can be 4 (sells to both) or 10 (sells only to Type 1).

$$p_B = 4 \Rightarrow \pi_B = 4(x+y)$$
$$p_B = 10 \Rightarrow \pi_B = 10x$$

Compare $10x$ vs $4(x+y)$.
$10x \geq 4x + 4y \Rightarrow 6x \geq 4y \Rightarrow 3x \geq 2y$, which holds since $x \geq y$. So set $p_B = 10$, $\pi_B = 10x$

Total profit from separate selling.
$\pi_{sep} = 10(x+y) + 10x = 20x + 10y$

(b) Highest Profit from Pure Bundling

> **Note**
>
> Bundling changes the product. The firm now sells a single "package" and customers evaluate it based on the sum of their valuations for the individual components.

Bundles by Consumer Types:
Type 1: $10 + 10 = 20$
Type 2: $12 + 4 = 16$

Market Coverage.
If $\rho_{AB} \leq 16$: Both types purchase
If $16 < \rho_{AB} \leq 20$: Only Type 1 purchases

Profit Functions.
Serve both: $\pi = 16(x+y)$
Serve Type 1 only: $\pi = 20x$

Profit Comparison.
Compare $16(x+y)$ vs $20x$:

$$16x + 16y \quad \text{vs} \quad 20x$$
$$16y \quad \text{vs} \quad 4x$$
$$4y \quad \text{vs} \quad x$$

Optimal Pricing Decision.
If $x \leq 4y$: Price at 16 to serve both types

If $x > 4y$: Price at 20 to serve only Type 1

Optimal Pure Bundling Price:

$$\rho_{AB} = \begin{cases} 16, & x \leq 4y \\ 20, & x > 4y \end{cases}$$

Pure Bundling Profit Function:

$$\pi^B = \begin{cases} 16(x+y) & x \leq 4y \\ 20x & x > 4y \end{cases}$$

Interpretation:

- When Type 2 consumers are relatively abundant ($x \leq 4y$), the firm prices low to capture both markets
- When Type 1 consumers dominate ($x > 4y$), the firm prices high to extract maximum value from the high-value segment

(c) Best Pricing Regime for All (x, y)

Individual Pricing: $\pi^{ID} = 20x + 10y$
Pure Bundling: $\pi^B = \max[16(x+y), 20x]$

Case 1: When $x \leq y$ is not applicable.

Case 2: When $x > y$

We compare the two bundling scenarios:

Subcase 2A: $\pi^B = 16(x+y)$
Compare with individual pricing:

$$20x + 10y > 16(x+y)$$

$$20x + 10y > 16x + 16y$$

$$20x - 16x > 16y - 10y$$

$$4x > 6y$$

$$x > \frac{3}{2}y$$

Thus.
If $x > \frac{3}{2}y$. Individual Pricing is better
If $x < \frac{3}{2}y$. Pure Bundling is better
If $x = \frac{3}{2}y$. Indifferent

Subcase 2B: $\pi^B = 20x$

Compare with individual pricing:

$$20x + 10y > 20x$$

This is always true since $10y > 0$ for $y \geq 1$

Thus:
Individual Pricing ($\pi^{ID} = 20x + 10y$) is always better than the $20x$ bundling option when $x > y$.

Note: In case there was any interest to the initial, handwritten (beta) solution, it can be found here. Do keep in mind that it's a mess and had plenty of wrong parts: https://drive.google.com/file/d/1wssjD7roK30XMTmwN4uqJKykkQ8FRBS/view?usp=sharing

It is not a 1 to 1 match considering that i polished this one, but the thought process is kind of the same.